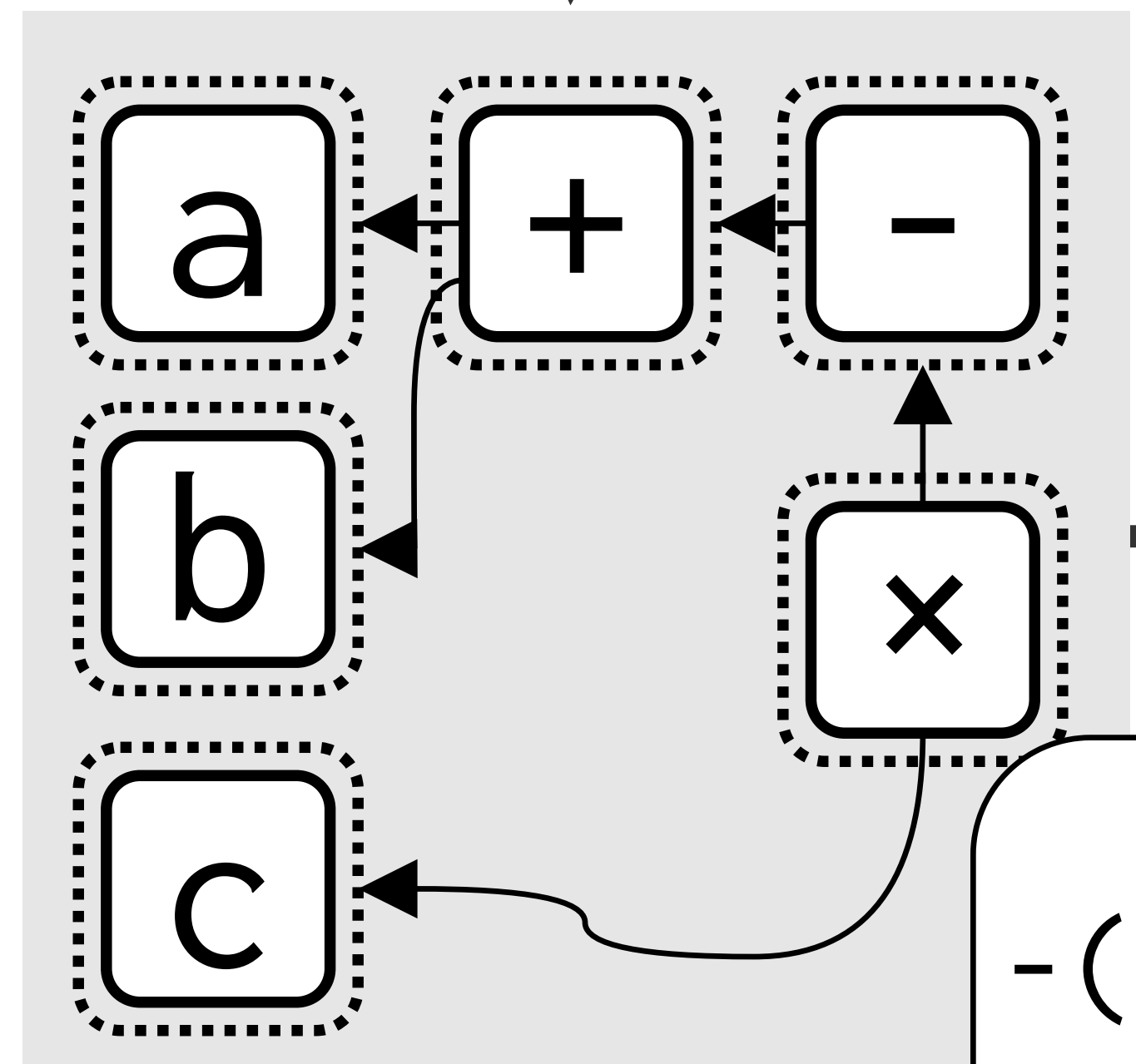


Operations (.mlir)
 $-(a+b) \times c$

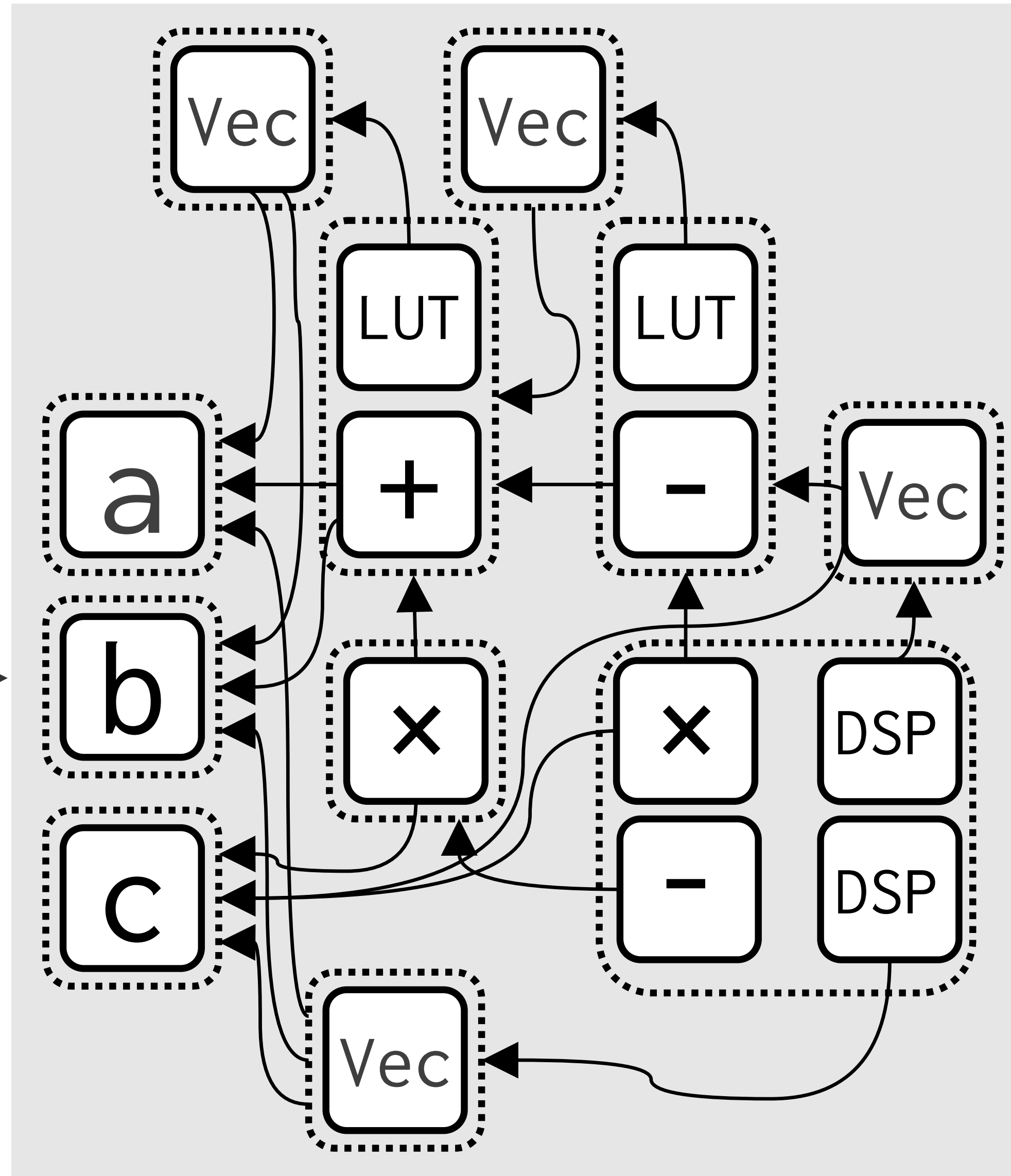


$\mathcal{G}_{init}$

Algebraic rules
 $-x \times y \rightsquigarrow -(x \times y)$

saturation phase

Implementation rules
 $-((A+D) \times B) \rightsquigarrow \text{DSP}(\text{Vec}(A, D, B))$
 $A \times B \rightsquigarrow \text{DSP}(\text{Vec}(A, B))$
 $A+B \rightsquigarrow \text{LUT}(\text{Vec}(A, B))$
 $-A \rightsquigarrow \text{LUT}(\text{Vec}(A))$



$\mathcal{G}_{saturate}$